

терпач обитан в кнелкы
120 нлсгос
New color Russia 5
5064807 000 "Слелелел"
Тел: 8301 120 нл, New color

Слелелел
слелел
2.1

2)

$$U_k = (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

$$M = \frac{U_k}{U_{k-1}} = \frac{(-1)^k \frac{x^{2k+1}}{(2k+1)!}}{(-1)^{k-1} \frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} =$$

$$= \frac{(-1) \cancel{(-1)^{k-1}} \frac{x^{2k+1}}{(2k+1)!}}{\cancel{(-1)^{k-1}} \frac{x^{2(k-1)+1}}{(2(k-1)+1)!}} = (-1) \cdot \frac{x^{2k+1}}{(2k+1)!}$$

$$\frac{(2(k-1)+1)!}{x^{2(k-1)+1}} = (-1) \frac{x^{2k} \cdot x \cdot (2(k-1)+1)!}{(2k+1)! \cdot \cancel{x^{2(k-1)}} \cdot \cancel{x^{2k}} \cdot \frac{1}{x^{1/2}}}$$

$$= (-1) \frac{x^2 (2k-1)!}{(2k+1)!} = (-1) \cdot \frac{x^2 (2k-1)!}{2k(2k+1)(2k-1)!}$$

$$(-1) \cdot \frac{x^2}{4k^2 + 2k}$$

$$M = - \frac{x^2}{4k^2 + 2k}$$

$$3) U_k = (-1)^k \cdot \frac{x^{2k}}{(2k)!}$$

$$M = \frac{U_k}{U_{k-1}} = \frac{(-1)^k \cdot \frac{x^{2k}}{(2k)!}}{(-1)^{k-1} \cdot \frac{x^{2(k-1)}}{(2(k-1))!}} =$$

$$= \frac{-1 \cdot \frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)!}} = \frac{-\frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)!}} =$$

$$= -\frac{\frac{x^{2k}}{(2k)!}}{\frac{x^{2k-2}}{(2k-2)!}} = -\frac{x^2 \cdot (2k-2)!}{(2k)!} =$$

$$= -\frac{x^2 \cdot (2k-2)!}{2k \cdot (2k-1) \cdot (2k-2)!} = -\frac{x^2}{2k \cdot (2k-1)} =$$

$$= -\frac{x^2}{4k^2 - 2k}$$

$$M = -\frac{x^2}{4k^2 - 2k}$$